

Linear Regression:

- Ma'ni: funksiyasi orqali class larga ajratadi (Sigmoid function).

K-cross Validation

• Cross Validation

- SML model qanday ko'rilgan dataset bilan ishlatiladi (ya'ni yoli yomon) baholash usuli.

• Turarlari:

- K-Fold cross validation
- Stratified K-Fold cross validation
- Leave-One-out cross validation (LOOCV)
- Leave-P-out cross validation

K-Fold split

- Datasetni k ta teng qismlarga ajratadi.
- Train $k-1$ folds
- Har safar 1 ta qism test, qolgan train sifatida ishlatiladi.
- Bu jarayon k marta takrorlanadi.
- Har bir natija-ning o'rtacha qiymati olinadi.

Caption

Amaliyotlari:

- Matematiklarni samarali foydalanadi.
- Modelning umumiy ishlashini aniqlashga yordam beradi.
- Overfittingni aniqlashga yordam beradi.

```
from sklearn.model_selection import cross_val_score
```

K-Fold cross validation.

```
from sklearn.model_selection import KFold, cross_val_score
```

```
Kf = KFold(n_splits=5, shuffle=True,  
           random_state=42)
```

```
scores = cross_val_score(lr, X, y, cv=Kf,  
                          scoring='r2')
```

```
print(scores)
```

```
import numpy as np  
np.mean(scores) → o'rtacha qiymati  
np.std(scores) → o'rtacha farq
```

mae hisoblash

```
from sklearn.metrics import mean_absolute_error  
mae = mean_absolute_error(y_true, y_pred,  
                           greater_is_better=False)
```

```
scores = cross_val_score(lr, X, y, cv=Kf,  
                          scoring=mae)
```

```
print(scores)
```

```
print(scores.mean())
```

